PingSheng Kao

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SUMMARY

Evaluation researcher and ML Data Scientist with 6+ years of experience measuring multimodal, LLM, agentic, and production ML systems. Builds simulated users, behavioral metrics, human-centered studies, scalable evaluation pipelines, and statistical failure analyses that feed directly into model development.

EXPERIENCE

ML Data Scientist - Apple July 2020 - Present
Santa Clara, California

- Built simulated-user and agent-as-judge systems to stress-test an open-ended multimodal conversational agent over long interactions, surfacing scenario-specific behavioral failures.
- Translated human-factors research into user-centered evaluation protocols and novel metrics for per-turn intent misalignment and false updates to an agent's inferred user state, estimating occurrence rates at scale.
- Built targeted scenario generators and evaluators for visual-memory and sycophancy weaknesses; systematic failure patterns changed subsequent development direction.
- Deployed and scaled LLM- and agent-as-judge evaluation on an existing internal batch platform, debugging throughput and rate limits, shipping versioned evaluator updates, and reporting results through dashboards.
- Developed an auditable, reference-based LLM evaluator and expert/non-expert human-evaluation workflow deployed internally for performance reporting and failure discovery; reached 90% expert alignment on 40 examples.
- Developed embedding-driven failure discovery and leadership-facing model-release analysis, using clustering, OOD analysis, regression attribution, and online validation to feed systematic patterns back into model development.

RESEARCH CONTRIBUTION

GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks 2025
Experiment contributor (credited as Sam Kao) - accepted to ECCV - arXiv:2512.24592

- Contributed experiments to ECCV-accepted research on grounded, hypothesis-driven error-slice discovery for instance-level vision tasks.

ADDITIONAL EXPERIENCE

Algorithm and Architecture Intern - Deep Learning Compression - Ambarella June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

EDUCATION

M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor 2018 - 2020
B.S. in Electronic Engineering - National Taiwan University 2013 - 2017
GPA: 3.9/4.0 (B.S.)

SKILLS

Programming & Data: Python, SQL, C++, JavaScript, AWS Athena
Generative AI Evaluation: Agent-as-Judge, LLM-as-Judge, User Simulation, DSPy, Tool-Call Evaluation
Behavior & Alignment: Human Factors, Intent Alignment, Sycophancy, Human Evaluation, Gold Sets
Methods: Metric Design, Bayesian Analysis, Experiment Design, Embeddings, Clustering, OOD Analysis